

HARRISON DELECKI

hdelecki@stanford.edu | (612) 240-3399 | github.com/hdelecki

Education

Stanford University, PhD Candidate

Aeronautics and Astronautics, GPA: 4.0

Sep. 2022 – May 2025 (Exp.)

Stanford, CA

Stanford University, MS

Aeronautics and Astronautics, GPA: 4.0

Sep. 2020 – May 2022

Stanford, CA

Georgia Institute of Technology, BS

Aerospace Engineering, Computer Science Minor, GPA: 4.0

August 2015 – May 2020

Atlanta, GA

Research

Stanford Intelligent Systems Laboratory

Research Assistant

November 2020 – Present

Stanford, CA

- **Generative Models for Black-Box Safety Validation:** Investigating training algorithms for deep generative models to validate complex, high-dimensional autonomous systems with no knowledge of internal system.
- **Entropy-regularized POMDP Solvers:** Extending existing offline POMDP solvers with entropy-regularized objectives to enable more robust planning under model uncertainty.
- **Model-Based Safety Validation:** Developing algorithms based on Bayesian inference and generative modeling to validate autonomous systems under very high dimensional disturbances.
- **Adaptive Stress Testing of Industry AVs:** Formulated reinforcement learning-based methods to find failures in real-world LIDAR-based object detection, tracking and trajectory prediction modules for autonomous driving.
- **Multi-Objective Planning for Subterranean Exploration:** Collaborated with NASA JPL's team CoStar to develop POMDP-based global planning solution for robotic subterranean exploration in the DARPA SubT Challenge.

Georgia Tech Space Systems Design Laboratory

Undergraduate Researcher

August 2018 – May 2020

Atlanta, GA

- **TARGIT CubeSat ADCS:** Led development of attitude determination and control system of university sponsored satellite.

Industry Experience

Motional

Safety Engineering Intern

June 2022 – September 2022

Boston MA

- Integrated reinforcement learning-based safety validation with industry AV stack and simulation to enable efficient safety analysis.

The Aerospace Corporation

Navigational Systems Intern

June 2020 – September 2020

El Segundo, CA

- Led team of undergraduates in investigation of reinforcement learning for packet routing in orbital networks.
- Implemented fast satellite selection algorithm for near-real-time GNSS performance assessment platform.

MIT Lincoln Laboratory

Summer Research Assistant

May 2018 – August 2018, May 2019– August 2019

Lexington, MA

- Developed software in MATLAB for tactical space situational awareness applications (SSA).
- Integrated real-time software into larger SSA decision making architecture.

MIT Lincoln Laboratory

Student Co-Op

June 2017 – December 2017

Lexington, MA

- Performed conceptual satellite design including mass/power budget, mechanical layout, and orbital analysis.
- Conducted design and analysis of metal 3D printed propellant management device for small satellites.

Skills

Programming Languages: Julia, Python, C/C++, Matlab, HTML/CSS, JavaScript, SQL

Technologies/Frameworks: Linux, ROS, GitHub, AWS, PyTorch/TensorFlow

Awards and Fellowships

National Science Foundation Graduate Research Fellowship

2022

Donnell Dutton Outstanding Senior in Georgia Tech Aerospace: For highest GPA in graduating class.

2020

Mathew Isakowitz Fellowship: For outstanding students interested in commercial space industry.

2020

Teaching and Mentorship

Teaching Assistant for AA228 and AA222. Wrote programming assignments, held weekly office hours for 450 students. **2022, 2023**

Research Mentor for Jacob Frausto (neural surrogates for validation), Yash Kadadi (transfer learning), Ellie Prince (decomposition for scalable decision-making), and Selena Sun (perception validation), including weekly meetings.